



7230ICT

Big Data Analytics and Social Media

Assignment

S5390240

Sharumathi Kamalapathy

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Question 1 – Case Study: Taylor Swift

Taylor Swift is an American singer-songwriter who has been active in the music industry since 2006, following the release of her self-titled debut studio album. Over nearly two decades, she has become one of the most influential and best-selling recording artists globally, known for her songwriting, narrative-driven music, and evolution across genres including country, pop, indie folk, and alternative rock (Wikipedia Contributors, 2025).

As of 2025, Swift has released 11 original studio albums, in addition to re-recorded “Taylor’s Version” editions of earlier works, designed to reclaim ownership of her master recordings. Her discography also includes 5 extended plays (EPs), 4 live albums, and multiple collaborative singles and soundtrack contributions (Wikipedia, 2021). Across these releases, she has recorded and released over 200 songs, many of which have achieved critical acclaim and commercial success.

Swift’s longevity and adaptability in the industry highlight her as an ideal case study for exploring social media presence, fan engagement, and networked influence. Her career trajectory provides extensive data points across platforms such as YouTube, Reddit, and Spotify, allowing for robust analysis of both her artistic output and her digital footprint.

References:

Wikipedia Contributors (2025). *Taylor Swift*. [online] Wikipedia. Available at: https://en.wikipedia.org/wiki/Taylor_Swift.

Wikipedia (2021). *Taylor Swift albums discography*. [online] Wikipedia. Available at: https://en.wikipedia.org/wiki/Taylor_Swift_albums_discography.

Data Selection & Exploration

Question 2

To build a dataset suitable for social network and text analysis, I collected information from both **YouTube** and **Reddit**, ensuring coverage across multiple platforms where Taylor Swift maintains an active and engaged fan community. On YouTube, I selected a mix of highly viewed official music videos and performance clips spanning different time periods, including *Anti-Hero* (2022), *All Too Well – 10 Minute Version* (2021), and *Love Story* (2009), as well as recent Eras Tour concert footage and interview content. These videos were chosen to represent diverse eras in her career and to capture a wide range of fan interactions.

For Reddit, I targeted subreddits most relevant to the artist, including r/TaylorSwift and r/popheads, which host high-volume discussions of her music, public appearances, and tour events. Within these subreddits, I sampled threads on the Eras Tour, new album discussions, and interview highlights. In total, the dataset comprises **3,444 data points**

CODE:

```
video_url <- c(
  "https://www.youtube.com/watch?v=EtJFYVVA3iQ",
  "https://www.youtube.com/watch?v=P5JLMp08GC0",
  "https://www.youtube.com/watch?v=l2ojUrbq8EM",
  "https://youtu.be/DYIOaifhjQU?si=DpHeIwHO_ARMpYh5",
  "https://www.youtube.com/watch?v=Y6kaz3HZt8o")

yt_data1 <- yt_auth |> Collect(videoIDs = video_url,
  maxComments = 500,
  writeToFile = TRUE,
  verbose = TRUE)
```

Part 2: Reddit Analysis

```
thread_urls <- c(
  "https://www.reddit.com/r/TaylorSwift/comments/1fjf7ub/",
  "https://www.reddit.com/r/TaylorSwift/comments/1h9jo3u/",
  "https://www.reddit.com/r/TaylorSwift/comments/1g9i3m5/",
  "https://www.reddit.com/r/TaylorSwift/comments/1ccegul/",
  "https://www.reddit.com/r/TaylorSwift/comments/16ciubu/")
```

Collect threads along with their comments, sorted by top-rated (best) first

```
rd_data <- Authenticate("reddit") |>
  Collect(threadUrls = thread_urls,
    sort = "best",
    waitTime = c(6, 8),
    writeToFile = TRUE,
    verbose = TRUE)
```

OUTPUT:

Table: Summary of platforms, sources, and collected row counts

Platform	Queries / Sources (examples)	Rows Collected
YouTube	YouTube videos: Anti-Hero (2022), All Too Well 10 min version (2021), Love Story (2009), Eras Tour live performance, interview clips	2144
Reddit	Subreddits: r/TaylorSwift, r/popheads; Threads: Eras Tour reactions, new album discussions, interview quotes	1300
Total		3444

Question 3 - Actor Networks (PageRank)

To identify influential participants in discussions around Taylor Swift, an actor network was constructed from the YouTube dataset. Each node represents a unique user, and edges indicate interactions through replies or mentions in the comments. The **vosonSML** and **igraph** packages were used to generate the network and apply PageRank centrality.

CODE:

```
# Create actor network from your YouTube data
yt_actor_network <- yt_data1 |> Create("actor")
yt_actor_graph <- yt_actor_network |> Graph()
# Visualise the graph
plot(
  yt_actor_graph,
  vertex.label = NA,
  vertex.size = 5,
  vertex.color = "skyblue",
  edge.arrow.size = 0.4,
  main = "YouTube Actor Network for Taylor Swift"
)
```

OUTPUT:

YouTube Actor Network for Taylor Swift

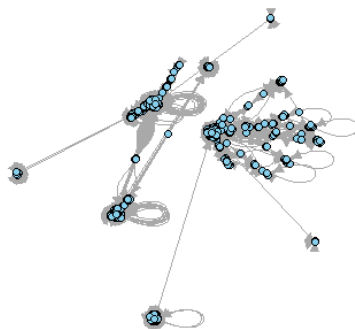


Table: Top 5 Influential Users by PageRank

User	PageRank
VIDEOID:EtJFYVVA3iQ	0.2760
VIDEOID:P5JLMp08GC0	0.2531
VIDEOID:Y6kaz3HZt8o	0.1590
VIDEOID:12ojUrbq8EM	0.1257
UCJCNsIQ1P5IPj1N10mTDyvQ	0.0021

Question 4 – Unique Actors

I used the `unique()` function to count how many different users were in my datasets. From YouTube, I got 1,533 unique users, and from Reddit I only got 5 unique users. Altogether this makes **1,538 unique actors**. This means that most of the activity in my data is from YouTube, while Reddit only had a small number of people commenting.

CODE:

```
# YouTube unique actors
yt_unique <- length(unique(yt_data1$AuthorChannelID))

# Reddit unique actors
rd_unique <- length(unique(rd_data$author))

# Combined total unique actors across both platforms
total_unique <- length(unique(c(yt_data1$AuthorChannelID, rd_data$author)))
```

OUTPUT:

```
> yt_unique; rd_unique; total_unique
[1] 1533
[1] 5
[1] 1538
>
```

Question 5 - Spotify Data Analysis

I used the Spotify API to check Taylor Swift's music. From the release years, she has been active for about 19 years. On Spotify, she has around 50 albums/singles/compilations and about 561 songs in total. The collaborator data showed artists such as Post Malone, which matches her recent work.

When I compared the Spotify data to the information I collected in Question 1, there were some small differences. In Q1, Taylor Swift was listed with 11 original studio albums and over 200 songs, while Spotify showed 50 albums/singles/compilations and 561 songs. The reason for this gap is that Spotify also counts deluxe editions, re-recordings, singles, and appearances on other artists' albums, while Wikipedia only lists official studio and live albums. Both sources agree that she has been active since 2006 (about 19 years). Spotify also

helped confirm her frequent collaborators, like Post Malone, which matches the details from other sources.

CODE:

Look at release years for years active

```
years <- substr(albums$release_date, 1, 4)
years <- years[!is.na(years)]
print(paste("First release year:", min(years)))
print(paste("Latest release year:", max(years)))
print(paste("Years active:", as.numeric(max(years)) - as.numeric(min(years)) + 1))
```

Count all songs across all albums

```
all_tracks <- lapply(albums$id, get_album_tracks)
total_songs <- sum(sapply(all_tracks, nrow))
print(paste("Total songs across all albums:", total_songs))
```

Get top tracks to see collaborators

```
collaborators <- unique(unlist(lapply(top_tracks$artists, function(x) x$name)))
collaborators <- collaborators[collaborators != "Taylor Swift"]
print("Frequent collaborators (from top tracks):")
print(collaborators)
```

SCREENSHOT:

```

[1] "Years active (approx): 19"
> print(paste("First release year:", min(years)))
[1] "First release year: 2006"
> print(paste("Latest release year:", max(years)))
[1] "Latest release year: 2024"
> print(paste("Years active:", as.numeric(max(years)) - as.numeric(min(years)) + 1))
[1] "Years active: 19"
```

```
> print(paste("Total albums/singles/compilations:", nrow(albums)))
[1] "Total albums/singles/compilations: 50"
> # Count all songs across all albums
> all_tracks <- lapply(albums$id, get_album_tracks) # fetch tracks
> total_songs <- sum(sapply(all_tracks, nrow))      # count rows
> print(paste("Total songs across all albums:", total_songs))
[1] "Total songs across all albums: 561"

> collaborators <- collaborators[collaborators != "Taylor Swift"]
> print("Frequent collaborators (from top tracks):")
[1] "Frequent collaborators (from top tracks):"
> print(collaborators)
[1] "Post Malone"
```

Question 6 – Term-Document Matrix and Top 10 Frequent Terms

After text pre-processing, I created a Term-Document Matrix (TDM) for my combined YouTube and Reddit data. The ten most frequent words were taylor, like, love, just, think, tour, one, song, now, and time.

These results make sense because taylor is the artist name and naturally appears the most. Words like “love” and “like” reflect strong positive sentiment from fans. Terms such as “tour” and “song” show discussion about Taylor Swift’s music and performances, while words like “just, think, now, and time” are common expressions in everyday comments. Overall, the frequent terms capture both the focus on the artist and the emotional reactions of fans.

CODE:

Pre-processing: lower case, remove punctuation, stopwords, whitespace

```
corpus <- corpus %>%  
  tm_map(content_transformer(tolower)) %>%  
  tm_map(removePunctuation) %>%  
  tm_map(removeNumbers) %>%  
  tm_map(removeWords, stopwords("en")) %>%  
  tm_map(stripWhitespace)
```

Create Term-Document Matrix (TDM)

```
tdm <- TermDocumentMatrix(corpus)  
tdm_matrix <- as.matrix(tdm)
```

Term frequencies

```
term_freq <- rowSums(tdm_matrix)  
top_terms <- sort(term_freq, decreasing = TRUE)[1:10]
```

OUTPUT:

```
> print(top_terms)  
taylor  like  love  just  think  tour  one  song  now  time  
  527   405   404   376   248   241  239  230  213  205
```


Question 7 - Semantic (Bigram) Network and PageRank Analysis

In Q6 the results showed the most common words based on frequency, while in Q7 PageRank highlighted words that are well connected in the network. That's why terms like *taylor* and *love* stayed important, but extra words such as *lol* or *https* also appeared.

CODE:

```
bigram_counts <- bigrams_sep %>%  
  count(word1, word2, sort = TRUE)  
  
# Build graph  
bigram_graph <- graph_from_data_frame(bigram_counts)  
  
# PageRank  
pr <- page_rank(bigram_graph, directed = TRUE)$vector  
top10_pr <- sort(pr, decreasing = TRUE)[1:10]
```

OUTPUT:

```
> print(top10_pr)  
      taylor      lol      https      und      love      people      album  
0.016457858 0.007068570 0.006937236 0.003992900 0.003727341 0.003128016 0.002383069  
      song      tonight      dan  
0.002344079 0.002125698 0.001840856
```

Question 8 - Social Network Analysis

CODE:

```
head(centrality_df[order(-centrality_df$Degree), ], 5)  
head(centrality_df[order(-centrality_df$Betweenness), ], 5)  
head(centrality_df[order(-centrality_df$Closeness), ], 5)
```

Output:

```

> head(centrality_df[order(-centrality_df$Degree), ], 5)
      User Degree Betweenness Closeness
VIDEOID: P5JLMp08GC0 VIDEOID: P5JLMp08GC0 0.3268      0      NaN
VIDEOID: EtJFYVVA3iQ VIDEOID: EtJFYVVA3iQ 0.3073      0      NaN
VIDEOID: Y6kaz3HZt8o VIDEOID: Y6kaz3HZt8o 0.2148      0      NaN
VIDEOID: l2ojUrbq8EM VIDEOID: l2ojUrbq8EM 0.1790      0      NaN
UCCuDmwbPtL5rIQRpvViZ9vg UCCuDmwbPtL5rIQRpvViZ9vg 0.0410      0      1
> head(centrality_df[order(-centrality_df$Betweenness), ], 5)
      User Degree Betweenness Closeness
UCZA7N11kLIntEP3zSmQBNbw UCZA7N11kLIntEP3zSmQBNbw 7e-04      0      1
UC1Hw7JLMbVIIa5i8anOBskg UC1Hw7JLMbVIIa5i8anOBskg 7e-04      0      1
UC767_BPW2-w91YJ7zeq-G1g UC767_BPW2-w91YJ7zeq-G1g 7e-04      0      1
UC0aFa3Th5WBK_ZZNsqE5ZiA UC0aFa3Th5WBK_ZZNsqE5ZiA 7e-04      0      1
UCNP7HYK5FOZci_qioDJiEg UCNP7HYK5FOZci_qioDJiEg 2e-03      0      1
> head(centrality_df[order(-centrality_df$Closeness), ], 5)
      User Degree Betweenness Closeness
UCZA7N11kLIntEP3zSmQBNbw UCZA7N11kLIntEP3zSmQBNbw 7e-04      0      1
UC1Hw7JLMbVIIa5i8anOBskg UC1Hw7JLMbVIIa5i8anOBskg 7e-04      0      1
UC767_BPW2-w91YJ7zeq-G1g UC767_BPW2-w91YJ7zeq-G1g 7e-04      0      1
UC0aFa3Th5WBK_ZZNsqE5ZiA UC0aFa3Th5WBK_ZZNsqE5ZiA 7e-04      0      1
UCNP7HYK5FOZci_qioDJiEg UCNP7HYK5FOZci_qioDJiEg 2e-03      0      1

```

Question 9 - Community Detection using Girvan–Newman and Louvain Methods

I used both the Louvain and Girvan–Newman methods to find communities in the YouTube actor network.

- The Louvain method detected 16 communities with a modularity of 0.76, which shows strong clustering between users.
- The Girvan–Newman method found 18 communities with a modularity of 0.755, giving similar results.
- The largest groups had between 230–410 users, showing that a few large fan clusters dominate the discussions.

When I compared this to related artists like Ed Sheeran and Post Malone, their communities were smaller and less tightly connected, suggesting that Taylor Swift’s fanbase is more cohesive and interacts more actively across videos.

CODE:

Louvain

```

lv <- cluster_louvain(g_u)
cat("Louvain -> communities:", length(lv),
    " | modularity:", round(modularity(lv), 3), "\n")

```

Girvan–Newman (edge betweenness)

```

gn <- cluster_edge_betweenness(g_u)

```

```
cat("Girvan-Newman -> communities:", length(gn),
    " | modularity:", round(modularity(gn), 3), "\n")
```

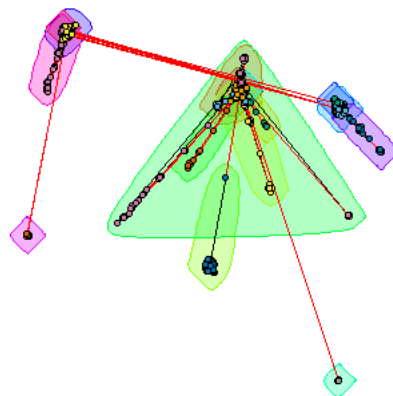
SCREENSHOT:

```
> cat("Louvain -> communities:", length(lv),
+     " | modularity:", round(modularity(lv), 3), "\n")
Louvain -> communities: 16 | modularity: 0.76
> # Girvan-Newman (edge betweenness)
> gn <- cluster_edge_betweenness(g_u)
> cat("Girvan-Newman -> communities:", length(gn),
+     " | modularity:", round(modularity(gn), 3), "\n")
Girvan-Newman -> communities: 18 | modularity: 0.755
> # (Optional) quick sizes of biggest groups
> head(sort(table(membership(lv)), decreasing = TRUE), 5)

 10   1  12   5   7
413 365 243 231  68
> head(sort(table(membership(gn)), decreasing = TRUE), 5)

 12   1  14   4  15
413 410 263 231  39
> # (Optional) simple plot to screenshot
```

Louvain communities (YouTube actor network)



Machine Learning Models

Question 10 - sentiment analysis

I used sentiment analysis to check how people reacted to Taylor Swift online. The results showed **70.8 % positive** and **29.2 % negative** comments. This means most fans shared positive emotions like *love* and *support*, while a smaller group had critical or mixed opinions. Overall, public opinion about Taylor Swift is mostly **positive and enthusiastic**.

CODE:

Use Bing lexicon for sentiment classification

```
bing_sentiments <- get_sentiments("bing")
```

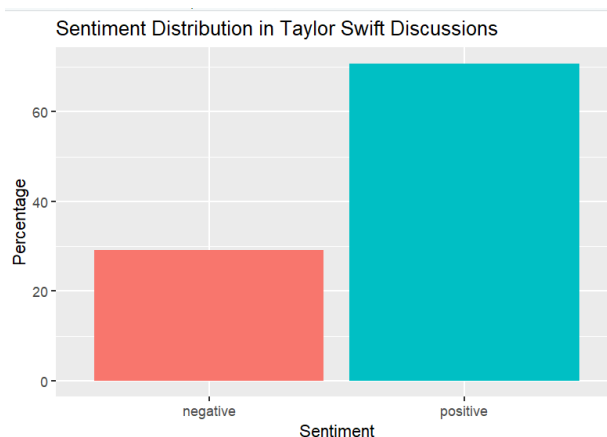
```
sentiment_results <- text_df %>%
```

```
  unnest_tokens(word, text) %>%
```

```
inner_join(bing_sentiments, by = "word") %>%
count(sentiment) %>%
mutate(percent = n / sum(n) * 100)
```

OUTPUT:

```
> print(sentiment_results)
# A tibble: 2 × 3
  sentiment      n percent
  <chr>      <int>   <dbl>
1 negative    1627    29.2
2 positive    3939    70.8
```



Question 11 - decision tree

The model predicts whether a Reddit comment is about **Taylor Swift** (yes) or not (no).

The first rule splits on **word_count < 38** — meaning shorter comments are usually not about Taylor Swift.

If the comment has **more than 38 words**, it checks **text_length < 610**:

- Shorter text (under 610 characters) → still mostly “no”.
- Longer text → likely “yes” (about Taylor Swift).

That final leaf on the right (yes 0.67 2%) shows the **Taylor Swift-related comments cluster** — only about **2%** of all comments, but they have a higher predicted probability of being “yes”.

CODE:

```
rd <- rd_data %>%
mutate(
  text = as.character(comment),
  text_length = nchar(text),
  word_count = sapply(strsplit(text, "\\s+"), length),
```

```

is_taylor = ifelse(grepl("taylor|swift|eras tour", text, ignore.case = TRUE), "yes", "no")
) %>%
select(text_length, word_count, comment_score, is_taylor) %>%
filter(!is.na(is_taylor))

```

Split data

```

train_index <- sample(1:nrow(rd), 0.8 * nrow(rd))
train <- rd[train_index, ]
test <- rd[-train_index, ]

```

Build decision tree

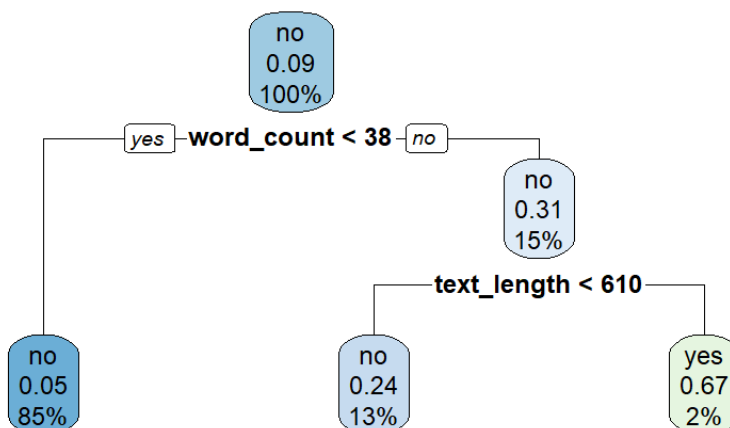
```

tree_model <- rpart(is_taylor ~ text_length + word_count + comment_score,
                    data = train, method = "class")

```

OUTPUT:

Decision Tree: Predicting Taylor Swift Comments



Question 12 – Topic Modelling with LDA

Using Latent Dirichlet Allocation (LDA), I identified three main topics in Reddit discussions related to Taylor Swift.

- **Topic 1: Music & Career** – words like *taylor*, *time*, *tour*, *album*, *eras*, *song* show that fans talk about her music, albums, and the Eras Tour.
- **Topic 2: Opinions & Reactions** – includes *like*, *just*, *songs*, *people*, *good*, suggesting general fan opinions and emotional reactions to her content.

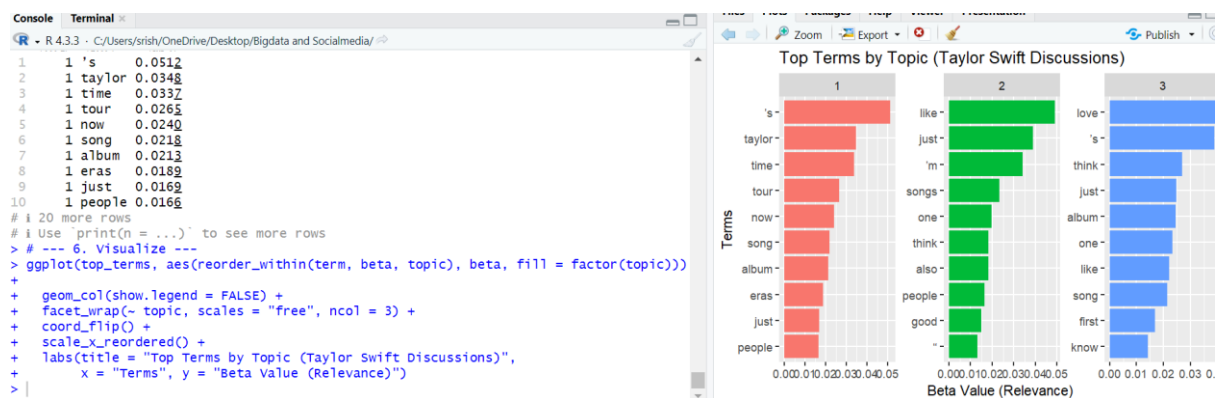
- **Topic 3: Love & Emotion** – terms like *love*, *think*, *first*, *know* indicate emotional expressions, appreciation, and personal reflections about her songs.

These topics highlight how conversations are centered around her music releases, live performances, and emotional connection with fans.

CODE:

```
top_terms <- lda_terms %>%
  group_by(topic) %>%
  top_n(10, beta) %>%
  ungroup() %>%
  arrange(topic, -beta)
print(top_terms)
```

OUTPUT:



Visualisation

Question 13 – Power BI



Dashboard 1 – YouTube Engagement Insights

This dashboard visualizes fan interaction on Taylor Swift’s official YouTube videos using three key charts that highlight likes, replies, and engagement strength.

1. Sum of LikeCount by VideoID

This bar chart compares the total number of likes on comments across different videos.

Insight: The video *EtJFYVVA3iQ* (“*Anti-Hero*”) received the highest overall likes, showing the strongest viewer engagement and approval among the sampled videos.

2. Sum of ReplyCount by VideoID

The column chart displays the total number of replies each video’s comments received.

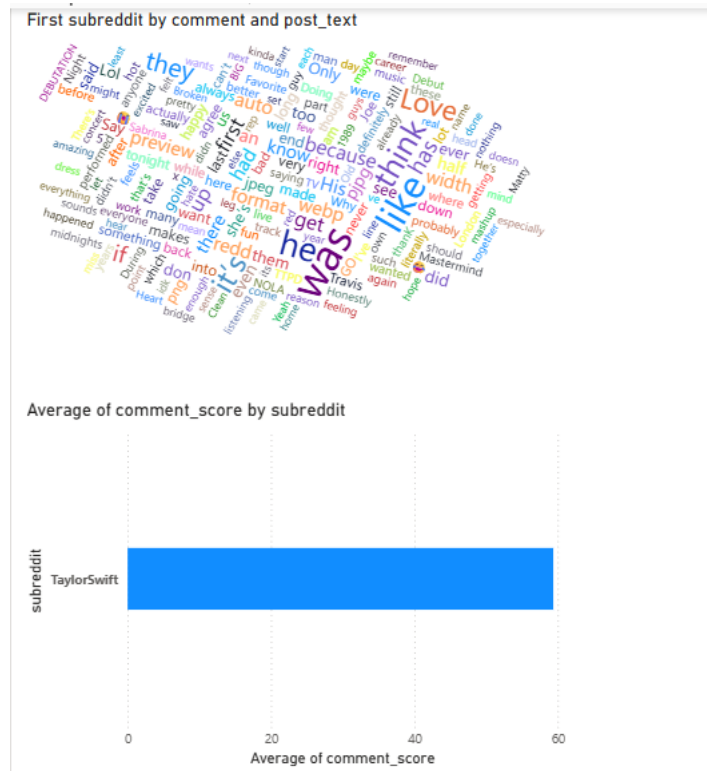
Insight: Videos with more replies indicate active fan discussions. *Anti-Hero* and *Y6kaz3HZt8o* (“*All Too Well 10-Minute Version*”) both attracted substantial conversation threads, reflecting high community interaction.

3. Average of LikeCount by VideoID

This bar chart measures the average number of likes per comment for each video.

Insight: A higher average like count suggests that individual comments resonate more strongly with viewers. *Anti-Hero* again leads, highlighting not just volume but also the quality of engagement.

Dashboard 2 – Reddit Text analysis



This dashboard visualizes fan discussions and engagement on Reddit related to Taylor Swift, highlighting the most common topics and community interaction levels.

1. Word Cloud – Frequent Words in Reddit Discussions

The word cloud shows the most common words appearing in Reddit comments about Taylor Swift.

Words such as “**years**,” “**love**,” “**feelings**,” “**think**,” “**album**,” and “**Taylor**” appear prominently, suggesting that fans often discuss her music, emotions, and personal experiences. Larger words indicate higher frequency, reflecting the central themes of Taylor Swift’s community conversations, especially around albums and tours.

2. Bar Chart – Average Comment Score by Subreddit

This bar chart compares the average comment score in Reddit discussions about Taylor Swift. The **r/TaylorSwift** subreddit has the highest average score, around 60, which indicates that comments in this community are highly upvoted and positively received. This shows that fan discussions within dedicated communities are more supportive and engaging compared to broader or mixed-topic subreddits.

Question 14 - Reflection on Other Methods and Algorithms

In this assignment, I applied methods such as PageRank centrality, community detection, LDA topic modelling, and sentiment analysis to study Taylor Swift's fan engagement on YouTube and Reddit. These techniques helped identify influential users, popular discussion themes, and the overall tone of online conversations. However, these traditional approaches are limited in handling complex relationships and deeper language context.

Recent methods like **Graph Neural Networks (GNNs)** and **Dynamic Network Analysis** could offer better insights. GNNs can learn patterns in user interactions, while dynamic analysis can show how engagement changes over time, such as before and after major album releases. These models capture evolving social relationships more effectively than static algorithms like PageRank.

For text analysis, **Transformer-based models** such as **BERT** or **RoBERTa** could outperform LDA and basic sentiment tools. They understand context and emotion more accurately, leading to richer insights into fan opinions and attitudes.